

Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) — YEAR 9 SEMESTER 1

COMPLETE COURSE OVERVIEW

ACADEMIC CALENDAR

Start: End of August

End: Chinese New Year (~18 weeks)

Lessons: 36 total (18 weeks x 2 lessons/week)

Lesson Duration: 40 minutes

STUDENT PROFILE

- EAL (English as Additional Language): Heavy bilingual scaffolding
- Mixed Ability: Differentiated worksheets (core/extension)
- Prior Knowledge: Minimal computing background assumed
- Programming: Introduced lightly in Weeks 16-17; full programming in Year 10

EXAM BOARD: CIE IGCSE 0478

Paper 1 (Theory): 1hr 45min, 75 marks, 50%

Paper 2 (Problem-solving): 1hr 45min, 75 marks, 50%

No coursework/NEA — purely exam-based

Calculators NOT permitted

PROGRAMMING STRATEGY

Year 9: Theory-focused with light Python exposure (Weeks 16-17)

Year 10: Full programming curriculum (selection, iteration, arrays, file handling)

Rationale: 2 lessons/week insufficient for deep programming + full theory

ASSESSMENT CALENDAR

Binary/Hex Quiz | Week 3 | 15 min | In-class

Data Representation Test | Week 6 | 30 min | End of unit

Hardware Quiz | Week 10 | 15 min | In-class

Software & Networks Quiz | Week 14 | 15 min | In-class

End-of-Semester Mock | Week 18 | 1 hour | Paper 1 style

RESOURCE REQUIREMENTS

- Mini-whiteboards (1 per student)
- A4 place-value cards (8 sets)

- Grid paper for binary addition
- 10x10 grid paper for pixel activity
- Bilingual glossary printouts
- Laminated place-value desk mats
- Physical "packet" cards for simulation
- Thonny IDE installed (Week 13)

CURRICULUM MAP — 18 WEEKS

**Week 1 | Binary & Denary (intro, place value) | Binary
& Denary (conversion practice) | 1.1.1**

**Week 2 | Binary Addition (rules, carry) | Binary
Addition (overflow, exam questions) | 1.1.2**

**Week 3 | Hexadecimal (intro, conversion) |
Hexadecimal (exam practice, why hex) | 1.1.3**

Week 4 Text Representation (ASCII, Unicode) (file size calc, Chinese characters)	Text 1.1.4
Week 5 Images (pixels, resolution, colour depth) Images (file size calc, metadata, exam questions)	 1.1.5
Week 6 Sound (sampling, bit depth) Compression (lossy vs lossless, RLE)	 1.1.6
Week 7 Data Transmission (packet switching) Packet switching (simulation, exam questions)	 1.2.1
Week 8 Error Detection (parity, checksum) Encryption (Caesar cipher, modern encryption)	 1.2.2
Week 9 CPU: FDE Cycle (intro, registers) (trace tables, exam questions)	FDE Cycle 1.3.1
Week 10 Memory: RAM, ROM, Cache Memory (hierarchy, virtual memory, exam questions)	 1.3.2
Week 11 Storage (magnetic, optical, solid-state) Devices & Sensors (embedded systems)	I/O 1.3.3
Week 12 Software: OS Functions & IDEs (compiler, interpreter, Thonny)	Translators 1.4.1
Week 13 Internet: WWW, URLs, DNS Networks: HTTP/HTTPS, IP addresses	 1.5.1
Week 14 Cybersecurity: Threats Cybersecurity: Protections (firewalls, 2FA)	 1.5.2
Week 15 Emerging Tech: Digital Currency Emerging Tech: Blockchain, AI, Robotics	 1.6.1

Week 16 | Programming Intro: Python Basics |
Programming: Variables, Input/Output | 2.1.1

Week 17 | Revision Clinic: Binary & Hardware |
Revision Clinic: Networks & Software | REVISION

Week 18 | Mock Exam (Paper 1 style, 1hr) | Mock
Review & Gap Analysis | ASSESSMENT

STANDARD LESSON TEMPLATE (40 Minutes)

Time | **Activity** | **Purpose**

0-3 min | Do Now | Retrieval, settle, immediate engagement

3-12 min | Direct Instruction | New concept, visual, bilingual keywords

12-25 min | Guided Practice | Worked examples, pair work, AfL

25-35 min | Independent Practice | Worksheet, exam question, physical task

35-40 min | Plenary + Homework | Check understanding, set HW

EAL SUPPORT CHECKLIST (Every Lesson)

- ☐ Bilingual keywords displayed (English + Chinese)
- ☐ Visual aid or physical demonstration used
- ☐ Sentence starter provided for extended answers
- ☐ Pair work before individual work
- ☐ Mini-whiteboard check for whole-class AfL
- ☐ Homework bridges to next lesson

DIFFERENTIATION

Emerging | Modified worksheet, peer support, sentence frames, bilingual glossary

Developing | Standard worksheet, occasional hints, paired work

Secure | Extension tasks, exam questions, peer tutoring

BILINGUAL GLOSSARY — ENGLISH / CHINESE / PINYIN

UNIT 1: DATA REPRESENTATION

binary | 二进制 | er jin zhi

denary/decimal | 十进制 | shi jin zhi

hexadecimal | 十六进制 | shi liu jin zhi

bit | 位 | wei

byte | 字节 | zi jie

place value | 位值 | wei zhi

nibble	半字节	ban zi jie
ASCII	美国信息交换标准代码	一
Unicode	统一码	tong yi ma
character set	字符集	zi fu ji
encoding	编码	bian ma
pixel	像素	xiang su
resolution	分辨率	fen bian lv
colour depth	颜色深度	yan se shen du
bitmap	位图	wei tu
metadata	元数据	yuan shu ju
sampling rate	采样率	cai yang lv
bit depth	位深度	wei shen du
analogue	模拟	mo ni
digital	数字	shu zi
compression	压缩	ya suo
lossy	有损压缩	you sun ya suo
lossless	无损压缩	wu sun ya suo
run-length encoding	行程编码	xingcheng bian ma

UNIT 2: DATA TRANSMISSION

packet switching	分组交换	fen zu jiao huan
data packet	数据包	shu ju bao
header	报头	bao tou
payload	有效载荷	you xiao zai he
error detection	错误检测	cuo wu jian ce
parity	奇偶校验	ji ou jiao yan
checksum	校验和	jiao yan he
encryption	加密	jia mi
decryption	解密	jie mi

UNIT 3: HARDWARE

CPU	中央处理器	zhong yang chu li qi
fetch-decode-execute	取指-译码-执行	qu zhi - yi ma - zhi xing
register	寄存器	ji cun qi
program counter	程序计数器	cheng xu ji shu qi
accumulator	累加器	lei jia qi
RAM	随机存取存储器	sui ji cun qu cun chu qi

ROM | 只读存储器 | zhi du cun chu qi
cache | 高速缓存 | gao su huan cun
virtual memory | 虚拟内存 | xu ni nei cun
sensor | 传感器 | chuan gan qi
actuator | 执行器 | zhi xing qi
embedded system | 嵌入式系统 | qian ru shi xi tong

UNIT 4: SOFTWARE

operating system | 操作系统 | cao zuo xi tong
compiler | 编译器 | bian yi qi
interpreter | 解释器 | jie shi qi
assembler | 汇编器 | hui bian qi
IDE | 集成开发环境 | ji cheng kai fa huan jing
interrupt | 中断 | zhong duan

UNIT 5: INTERNET & SECURITY

World Wide Web | 万维网 | wan wei wang
URL | 统一资源定位符 | tong yi zi yuan ding wei fu
DNS | 域名系统 | yu ming xi tong
HTTP | 超文本传输协议 | chao wen ben chuan shu xie yi
HTTPS | 安全超文本传输协议 | an quan chao wen ben chuan shu xie yi
IP address | IP 地址 | IP di zhi
router | 路由器 | lu you qi
firewall | 防火墙 | fang huo qiang
malware | 恶意软件 | e yi ruan jian
phishing | 网络钓鱼 | wang luo diao yu
two-factor auth | 双因素认证 | shuang yin su ren zheng

UNIT 6: EMERGING TECH

artificial intelligence | 人工智能 | ren gong zhi neng
machine learning | 机器学习 | ji qi xue xi
robotics | 机器人技术 | ji qi ren ji shu
blockchain | 区块链 | qu kuai lian
digital currency | 数字货币 | shu zi huo bi
automated system | 自动化系统 | zi dong hua xi tong

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